

Group 23 Project Protocol

Benjamin Sabu Zacharia Isabel Sandoval
Aditya Mangalampalli

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Experimental Protocol

Description

For this experiment we'll test how three factors (breathing, background video distraction, and time of day) affect reaction time in milliseconds. To do this, we'll use a repeated measures factorial design, blocking on the subjects. Each condition will have two levels; breathing (holding breath vs. breathing normally), distraction (playing a video in the background vs. omitting the video), and time of day (morning vs. evening). This results in a factorial design with eight unique combinations, with each of the 3 subjects going through all possible treatments multiple times. We expect a sample size of 48 observations to carry out this experiment.

Power Calculations and Results

We have defined our sample size to be the number of observations in our dataframe (hence a sample size of 48 after power calculations). Our estimate of σ^2 is 405.5, and we achieved this estimate conducting a 2x2x2 ANOVA analysis on our dry run after blocking on subjects, then observing the Mean Squared Residuals. We have multiple alternative hypothesis' for each factor and interaction, which follow the pattern : "Factor/Interaction has an effect on reaction time in milliseconds, any observable difference is NOT due to chance alone". We have decided on these alternative hypothesis' as it aligns with the goals of this project, which is to study the factors that effect a persons reaction time. We have attached the code for our power curves and the figures themselves below, under appendix A. Based on these power curves, we have concluded that a sample size of 48 is sufficient for the purposes of attaining enough power ($0.7 \leq 1 - \beta \leq 0.9$) for our experiment. We can detect relatively small effect sizes (less than 10 milliseconds in differences) with the experiments current power.

Protocol

The steps to conduct the experiment are listed below in below in order:

- Preparation
 - Environment: Sit in a room with consistent lighting and minimal background noise. Ensure you are at a table that allows for ergonomic use of your laptop.
 - Hardware Setup: Connect your laptop to a power source. Use a physical mouse if possible, rather than a trackpad, and stick to that choice for all trials
 - Software: Close all background applications and browser tabs except for the Human Benchmark Reaction Test.
- Randomization & Assignment
 - Use the provided R code to randomly assign treatments across each block.
- Execute the experiment
 - Look at your treatment assignments which consist of three factors (Breathing, Video, and Time of Day):
 - * Breathing Factor:
 - Normal: Breathe naturally for the duration of the test.
 - Holding Breath: Take a deep breath and hold it immediately before clicking "Start." Maintain the hold until the final click of that specific trial is completed.

- * Video Factor:
 - No Video (Control): Perform the test in silence.
 - Video (Distraction): Play a specific, high-stimulus video.
- * Time of Day Factor:
 - Perform “Morning” trials within 3 hours of waking up.
 - Perform “Evening” trials after 6:00 PM but before 10:00 PM.
- Data Collection and Measurement
 - Once adjusted to treatment for a specific trial as described above, tap start and click again as soon as the screen turns green. Record your time in milliseconds on a Google Spreadsheet.
 - Repeat this step until you complete all of your treatments as listed in the randomly assigned dataframe
 - Convert the spreadsheet to csv
- ANOVA analysis and conclusion
 - Once you’ve finished gathering all of you’re data, go to R studio and import your csv and assign it to ‘dat’.
 - Run all the cells in the provided code.
 - Based on the results of the ANOVA, conclude your analysis by deciding the accept or reject the multiple null hypothesis posed by the ANOVA analysis.

Preregistration

Once the data has been collected, we plan to run a 2x2x2 ANOVA on our data after blocking by the 3 subjects. Before this, we will set or significance threshold for all of our tests at 0.1. For each test: If the p-value for the main/interaction effect of the factor(s) is less than 0.05, then we will conclude that there is evidence that the factor(s) has an effect on response time in milliseconds.

Appendix 1: Dataframe of final assignments

```
n <- 48

# Create a balanced set of vectors to pass into data generating process
i <- sample(rep(1:2, each = n/2)) # 12 ones, 12 twos, shuffled
j <- sample(rep(1:2, each = n/2))
k <- sample(rep(1:2, each = n/2))
l <- sample(rep(1:3, each = n/3)) # 8 ones, 8 twos, 8 threes, shuffled

dat <- data.frame(
  breathing = factor(i),
  time = factor(j),
  video = factor(k),
  person = factor(l))

dat <- dat |>
  mutate(
    breathing = factor(breathing,
                      levels = c(1, 2),
                      labels = c("Normal", "Hold")),
    time      = factor(time,
                      levels = c(1, 2),
                      labels = c("Morning", "Evening")),
    video     = factor(video,
                      levels = c(1, 2),
                      labels = c("No", "Yes")),
    person    = factor(person,
                      levels = c(1, 2, 3),
                      labels = c("Aditya", "Benjamin", "Isabel"))
  )

print(dat)
```

	breathing	time	video	person
1	Normal	Morning	Yes	Isabel
2	Normal	Evening	Yes	Aditya
3	Hold	Morning	Yes	Aditya
4	Normal	Morning	No	Isabel
5	Hold	Evening	Yes	Isabel
6	Hold	Evening	No	Isabel

7	Normal Morning	No	Benjamin
8	Normal Morning	Yes	Isabel
9	Normal Evening	Yes	Benjamin
10	Hold Morning	No	Benjamin
11	Normal Evening	No	Isabel
12	Normal Morning	No	Benjamin
13	Hold Morning	No	Isabel
14	Hold Morning	No	Benjamin
15	Hold Evening	Yes	Aditya
16	Normal Evening	Yes	Aditya
17	Hold Morning	Yes	Benjamin
18	Hold Evening	Yes	Aditya
19	Hold Morning	Yes	Aditya
20	Normal Morning	No	Aditya
21	Hold Evening	No	Aditya
22	Hold Morning	Yes	Aditya
23	Hold Evening	No	Aditya
24	Normal Morning	No	Benjamin
25	Normal Evening	No	Isabel
26	Normal Morning	No	Benjamin
27	Normal Evening	No	Aditya
28	Hold Evening	Yes	Aditya
29	Normal Evening	No	Benjamin
30	Hold Evening	Yes	Benjamin
31	Hold Evening	No	Isabel
32	Normal Morning	No	Benjamin
33	Hold Morning	Yes	Benjamin
34	Normal Evening	Yes	Benjamin
35	Normal Evening	Yes	Isabel
36	Hold Evening	Yes	Isabel
37	Normal Morning	Yes	Benjamin
38	Normal Morning	No	Aditya
39	Hold Evening	No	Isabel
40	Normal Morning	Yes	Isabel
41	Hold Evening	Yes	Aditya
42	Hold Evening	Yes	Aditya
43	Normal Morning	Yes	Isabel
44	Hold Morning	No	Isabel
45	Normal Morning	No	Isabel
46	Normal Morning	No	Aditya
47	Hold Evening	Yes	Benjamin
48	Hold Evening	No	Benjamin

Appendix A: Anova Analysis (we used this to get our estimate of the variance; we also use this in our final analysis of experimental data)

ANOVA analysis

```
dat <- read.csv('dry_run.csv')
```

```
head(dat)
```

	Person	Time	Breathing	Video	Reaction_time_.ms.
1	Benjamin	Evening	Normal	No	211
2	Benjamin	Evening	Hold	No	182
3	Benjamin	Evening	Normal	Yes	215
4	Benjamin	Evening	Hold	Yes	195
5	Benjamin	Morning	Normal	No	236
6	Benjamin	Morning	Hold	No	220

```
fit<-lm(Reaction_time_.ms. ~Person + Breathing* Time* Video, data=dat)
aov_tab<-anova(fit)
aov_tab
```

Analysis of Variance Table

Response: Reaction_time_.ms.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Person	2	13592.3	6796.2	16.7619	0.0001925 ***
Breathing	1	630.4	630.4	1.5547	0.2328945
Time	1	155.0	155.0	0.3824	0.5462556
Video	1	1488.4	1488.4	3.6709	0.0760184 .
Breathing:Time	1	126.0	126.0	0.3109	0.5859539
Breathing:Video	1	12.0	12.0	0.0297	0.8656407
Time:Video	1	26.0	26.0	0.0642	0.8036168
Breathing:Time:Video	1	360.4	360.4	0.8888	0.3617824
Residuals	14	5676.3	405.5		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Hence, we use 405.5 as our estimate of σ^2

Appendix B: Power and Power Curves

Data Generating Process

```
generate_Y <- function(mu,
                        A, B, C, D,
                        AB, AC, BC,
                        ABC,
                        sigma_sq,
                        i, j, k, l) {

  n <- length(i)

  # irreducible error
  epsilon <- rnorm(n, mean = 0, sd = sqrt(sigma_sq))

  # These make up the main effects for our model
  main_effects <- mu + A[i] + B[j] + C[k] + D[l]

  # These are all the 2 way interactions for our model
  AB_effect <- AB[cbind(i, j)]
  AC_effect <- AC[cbind(i, k)]
  BC_effect <- BC[cbind(j, k)]

  # This is the 3 way interaction
  ABC_effect <- ABC[cbind(i, j, k)]

  # full model
  Y <- main_effects +
        AB_effect + AC_effect + BC_effect +
        ABC_effect +
        epsilon

  return(Y)
}
```

Parameters for Data Generation (including main and interaction effects)

```
mu <- 250

A <- c(8.5, -8.5)      # Breathing
B <- c(8.7, -8.7)      # Time
```

```

C <- c(8.3, -8.3) # Video
D <- c(-20, 10, 10) # Person (Block)

# 2-way interactions (2x2 matrices)

AB <- matrix(c(10, -10,
               -10, 10), nrow = 2)

AC <- matrix(c(9, -9,
               -9, 9), nrow = 2)

BC <- matrix(c(9.1, -9.1,
               -9.1, 9.1), nrow = 2)

# 3-way interaction (2x2x2 array)

ABC <- array(c(10, -10, -10, 10,
               -10, 10, 10, -10),
             dim = c(2, 2, 2))

```

Function to simulate one dataframe

```

one_sim <- function(mu, A, B, C, D, AB, AC, BC, ABC,
                   sigma_sq, n, target_effect) {

  # Create a balanced set of all treatment combinations
  treatments <- expand.grid(
    breathing = 1:2,
    time = 1:2,
    video = 1:2
  )
  full_design <- treatments[rep(1:nrow(treatments), length.out = n), ]

  # Shuffle the design so the 'Person' (block) is randomly assigned to these
  shuffled_design <- full_design[sample(1:n), ]

  i_sim <- shuffled_design$breathing
  j_sim <- shuffled_design$time
  k_sim <- shuffled_design$video
  l_sim <- sample(rep(1:3, each = n/3)) # Person block

```



```

Y <- generate_Y(mu, A, B, C, D, AB, AC, BC, ABC, sigma_sq,
               i_sim, j_sim, k_sim, l_sim)

dat <- data.frame(
  breathing = factor(i_sim),
  time = factor(j_sim),
  video = factor(k_sim),
  person = factor(l_sim),
  Y = Y
)

fit <- lm(Y ~ person + breathing * time * video, data = dat)

p_val <- anova(fit)[target_effect, "Pr(>F)"]
reject <- if(is.na(p_val)) 0 else as.integer(p_val < 0.05)

return(reject)
}

```

All Power Curves

```

n_values <- seq(from = 24, by = 12, length.out = 10)
factors_to_test <- c("breathing", "time", "video", "time:video",
                    "breathing:video", "breathing:time",
                    "breathing:time:video")

iter <- 100

# Run simulations across n and factors
power_data <- map_dfr(n_values, function(current_n) {

  # For each n, calculate power for each factor
  map_dfr(factors_to_test, function(f) {

    rejections <- replicate(iter, one_sim(mu, A, B, C, D, AB, AC,
                                          BC, ABC, 405.5, current_n, f))

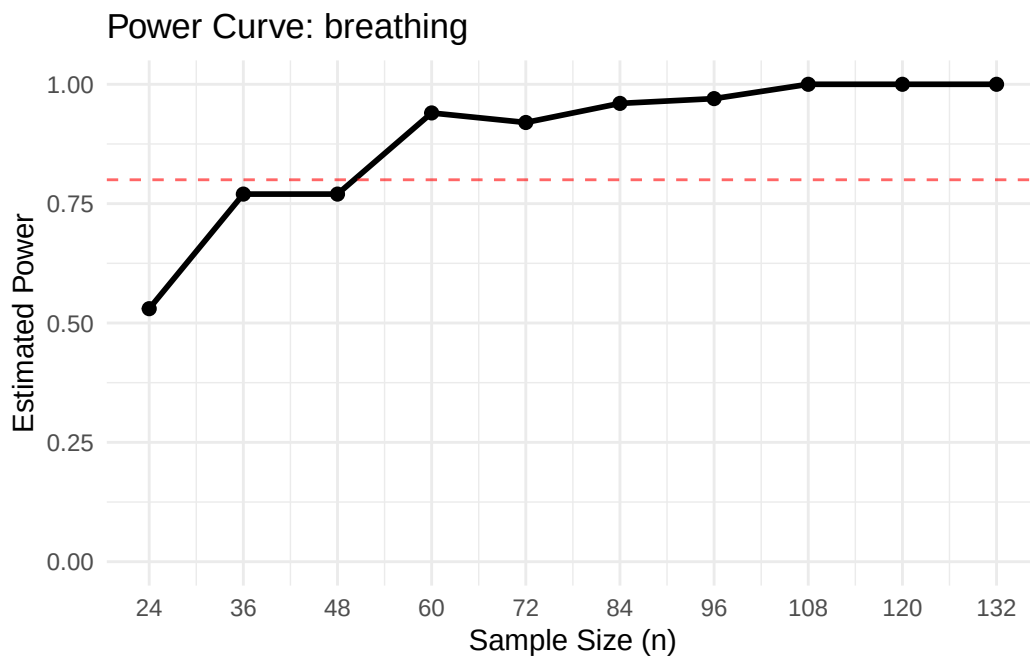
    data.frame(
      n = current_n,
      power = mean(rejections),
      factor = f
    )
  })
})

```

```
})  
})
```

```
selected_factor <- "breathing"
```

```
power_data |>  
  filter(factor == selected_factor) |>  
  ggplot(aes(x = n, y = power)) +  
  geom_line(color = "black", linewidth = 1) +  
  geom_point(color = "black", size = 2) +  
  geom_hline(yintercept = 0.8, linetype = "dashed",  
            color = "red", alpha = 0.6) +  
  scale_y_continuous(limits = c(0, 1)) +  
  scale_x_continuous(breaks = n_values) +  
  labs(  
    title = paste("Power Curve:", selected_factor),  
    x = "Sample Size (n)",  
    y = "Estimated Power"  
  ) +  
  theme_minimal()
```

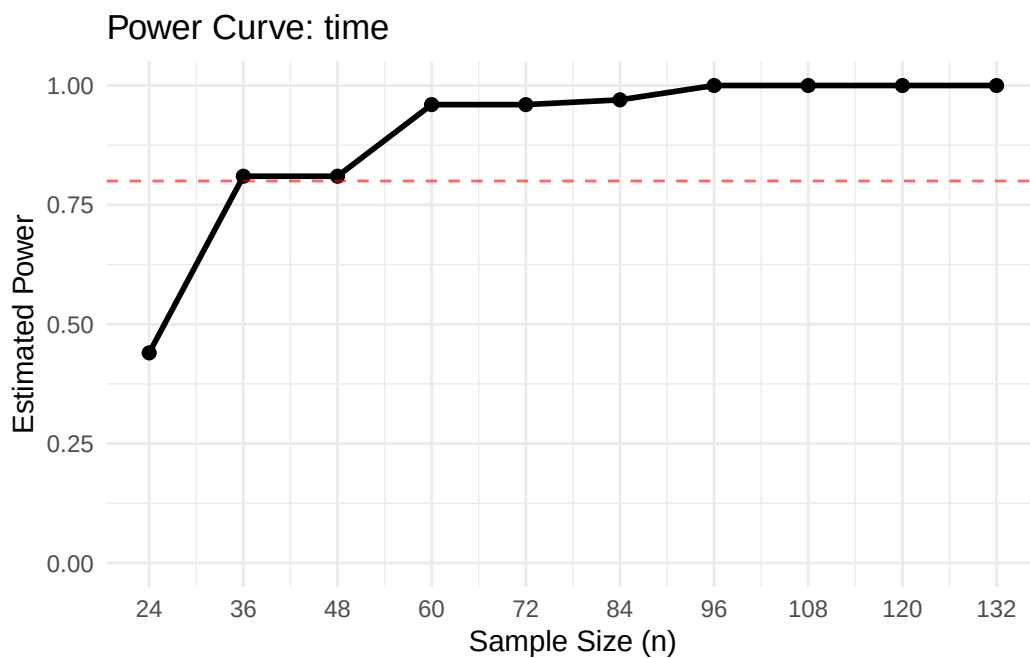


```

selected_factor <- "time"

power_data |>
  filter(factor == selected_factor) |>
  ggplot(aes(x = n, y = power)) +
    geom_line(color = "black", linewidth = 1) +
    geom_point(color = "black", size = 2) +
    geom_hline(yintercept = 0.8, linetype = "dashed",
              color = "red", alpha = 0.6) +
    scale_y_continuous(limits = c(0, 1)) +
    scale_x_continuous(breaks = n_values) +
    labs(
      title = paste("Power Curve:", selected_factor),
      x = "Sample Size (n)",
      y = "Estimated Power"
    ) +
    theme_minimal()

```



```

selected_factor <- "video"

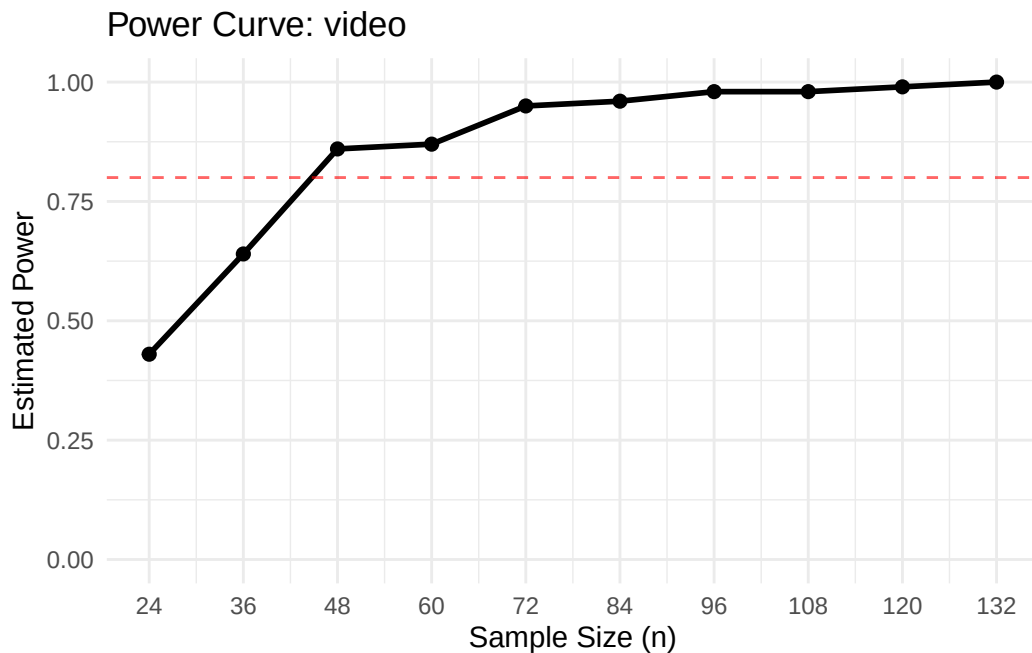
power_data |>
  filter(factor == selected_factor) |>

```

```

ggplot(aes(x = n, y = power)) +
  geom_line(color = "black", linewidth = 1) +
  geom_point(color = "black", size = 2) +
  geom_hline(yintercept = 0.8, linetype = "dashed",
             color = "red", alpha = 0.6) +
  scale_y_continuous(limits = c(0, 1)) +
  scale_x_continuous(breaks = n_values) +
  labs(
    title = paste("Power Curve:", selected_factor),
    x = "Sample Size (n)",
    y = "Estimated Power"
  ) +
  theme_minimal()

```



```

selected_factor <- "time:video"

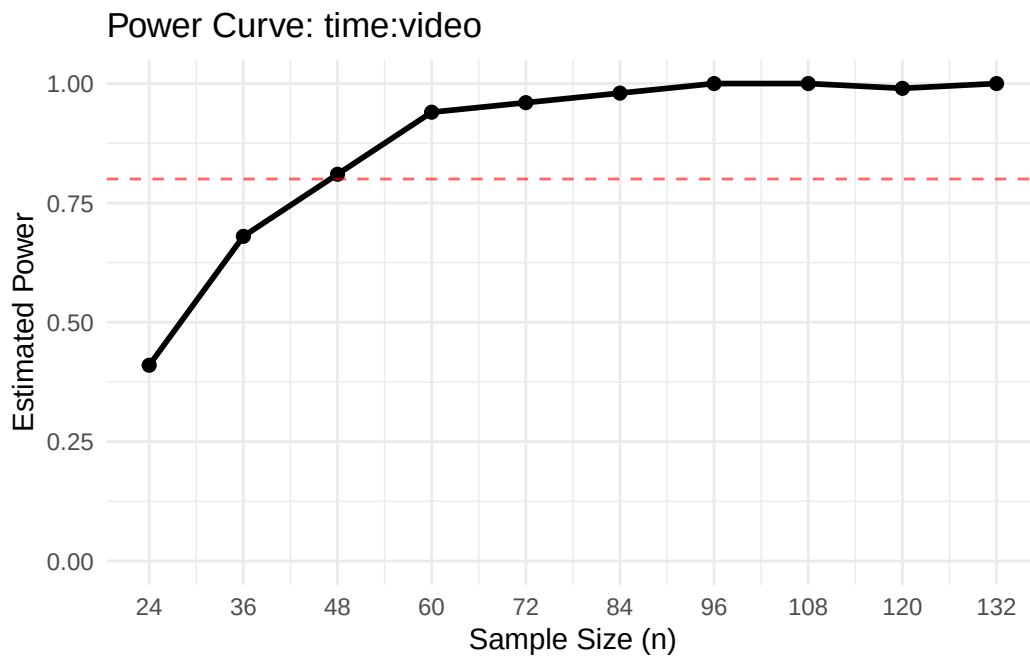
power_data |>
  filter(factor == selected_factor) |>
  ggplot(aes(x = n, y = power)) +
    geom_line(color = "black", linewidth = 1) +
    geom_point(color = "black", size = 2) +
    geom_hline(yintercept = 0.8, linetype = "dashed",

```

```

    color = "red", alpha = 0.6) +
scale_y_continuous(limits = c(0, 1)) +
scale_x_continuous(breaks = n_values) +
labs(
  title = paste("Power Curve:", selected_factor),
  x = "Sample Size (n)",
  y = "Estimated Power"
) +
theme_minimal()

```



```

selected_factor <- "breathing:video"

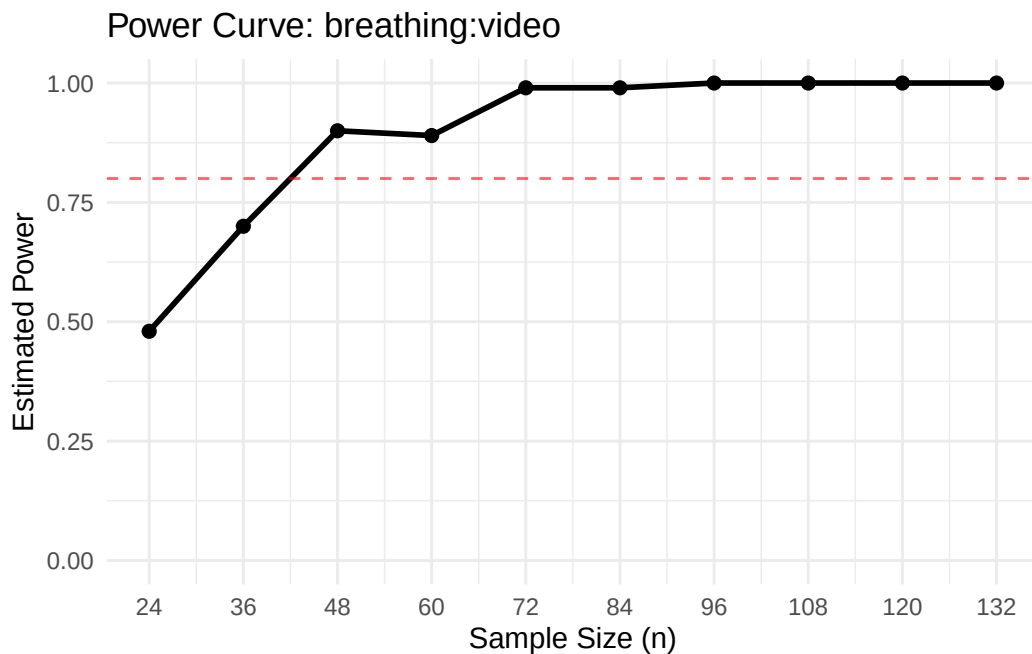
power_data |>
  filter(factor == selected_factor) |>
  ggplot(aes(x = n, y = power)) +
  geom_line(color = "black", linewidth = 1) +
  geom_point(color = "black", size = 2) +
  geom_hline(yintercept = 0.8, linetype = "dashed",
    color = "red", alpha = 0.6) +
  scale_y_continuous(limits = c(0, 1)) +
  scale_x_continuous(breaks = n_values) +
  labs(

```

```

    title = paste("Power Curve:", selected_factor),
    x = "Sample Size (n)",
    y = "Estimated Power"
  ) +
  theme_minimal()

```



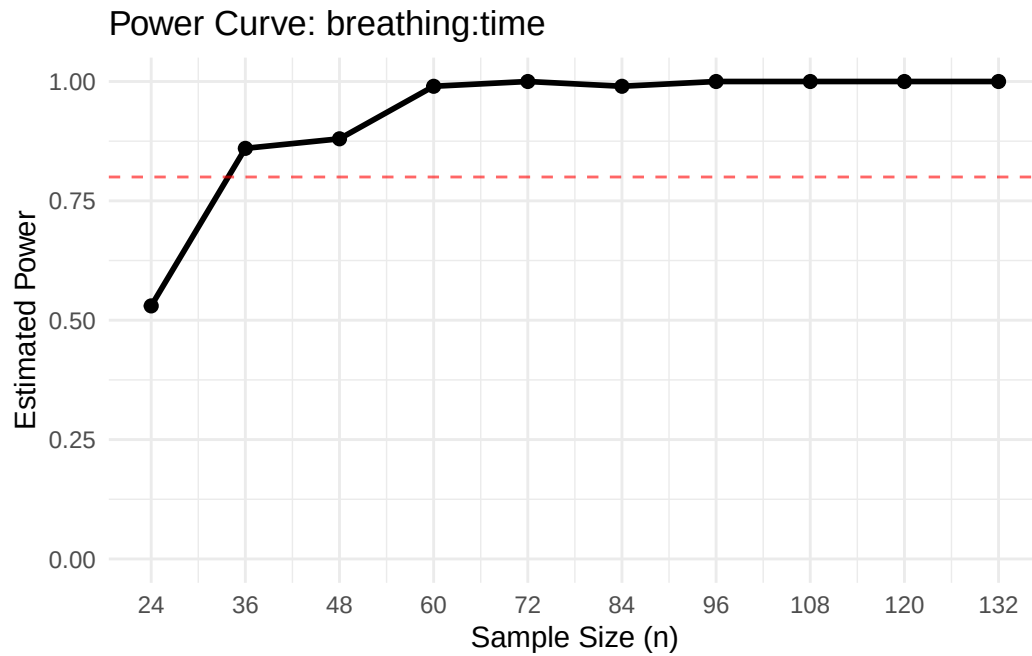
```

selected_factor <- "breathing:time"

power_data |>
  filter(factor == selected_factor) |>
  ggplot(aes(x = n, y = power)) +
  geom_line(color = "black", linewidth = 1) +
  geom_point(color = "black", size = 2) +
  geom_hline(yintercept = 0.8, linetype = "dashed",
             color = "red", alpha = 0.6) +
  scale_y_continuous(limits = c(0, 1)) +
  scale_x_continuous(breaks = n_values) +
  labs(
    title = paste("Power Curve:", selected_factor),
    x = "Sample Size (n)",
    y = "Estimated Power"
  ) +

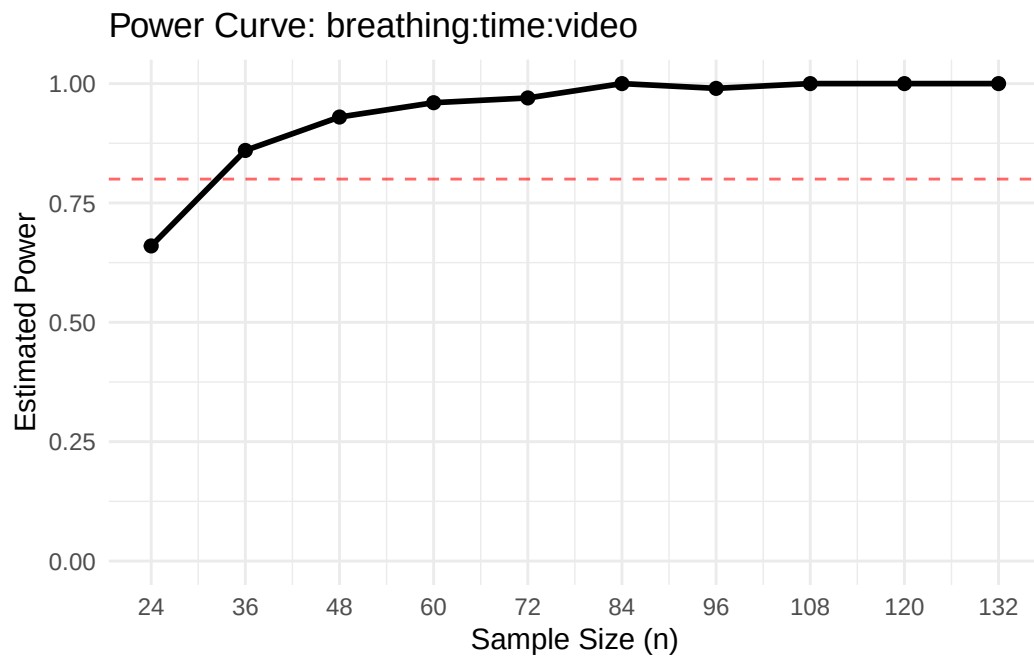
```

```
theme_minimal()
```



```
selected_factor <- "breathing:time:video"

power_data |>
  filter(factor == selected_factor) |>
  ggplot(aes(x = n, y = power)) +
  geom_line(color = "black", linewidth = 1) +
  geom_point(color = "black", size = 2) +
  geom_hline(yintercept = 0.8, linetype = "dashed",
             color = "red", alpha = 0.6) +
  scale_y_continuous(limits = c(0, 1)) +
  scale_x_continuous(breaks = n_values) +
  labs(
    title = paste("Power Curve:", selected_factor),
    x = "Sample Size (n)",
    y = "Estimated Power"
  ) +
  theme_minimal()
```



Appendix 2: Dry Run

We performed our Dry Run asynchronously, and we more or less followed the protocol exactly. One important change made was staying consistent in the hardware we used. We realized that conducting the test with a track pad often resulted in drastically higher reaction times as opposed to using a mouse.

```
dry_run <- read.csv("dry_run.csv")
dry_run
```

	Person	Time	Breathing	Video	Reaction_time_.ms.
1	Benjamin	Evening	Normal	No	211
2	Benjamin	Evening	Hold	No	182
3	Benjamin	Evening	Normal	Yes	215
4	Benjamin	Evening	Hold	Yes	195
5	Benjamin	Morning	Normal	No	236
6	Benjamin	Morning	Hold	No	220
7	Benjamin	Morning	Normal	Yes	232
8	Benjamin	Morning	Hold	Yes	228
9	Isabel	Evening	Normal	No	283
10	Isabel	Evening	Hold	No	269
11	Isabel	Evening	Normal	Yes	305
12	Isabel	Evening	Hold	Yes	263
13	Isabel	Morning	Normal	No	263
14	Isabel	Morning	Hold	No	231
15	Isabel	Morning	Normal	Yes	294
16	Isabel	Morning	Hold	Yes	273
17	Aditya	Evening	Normal	No	215
18	Aditya	Evening	Hold	No	260
19	Aditya	Evening	Normal	Yes	249
20	Aditya	Evening	Hold	Yes	275
21	Aditya	Morning	Normal	No	256
22	Aditya	Morning	Hold	No	232
23	Aditya	Morning	Normal	Yes	255
24	Aditya	Morning	Hold	Yes	263

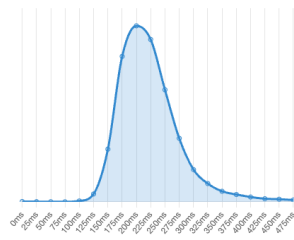
We use the human benchmark reaction time test. Attached below, is a sample run.



249 ms

Click to keep going

Statistics



About the test

This is a simple tool to measure your reaction time.

The average (median) reaction time is 273 milliseconds, according to [the data](#) collected so far.

In addition to measuring your reaction time, this test is affected by the latency of your computer and monitor. Using a fast computer and low latency / high framerate monitor will improve your score.

Scores in this test are faster than the [aim trainer](#) test, because you can react instantly without moving the cursor.

This is discussed in further detail on the [statistics](#) page. While an average human reaction time may fall between 200-250ms, your computer could have

Figure 1: Sample test Results